Vansh Chugh

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Research Interests: Audio effects representation learning, automatic mixing, real-time human–AI music co-creation

Education

B.A. in Data Science; Minors in Mathematics & Pop Music Sep 2023 – June 2027

Knox College | GPA: 4.0/4.0 *Galesburg, IL*

- **Senior Honors Thesis (in progress):** Real-Time Accompaniment Generation for Human–AI Co-Creation. Advisors: Benjamin Penwell, Ole J. Forsberg
- **Junior Honors Thesis** (sole junior admitted to Honors Program): Context-Aware Representation Learning for Mix Compatibility Estimation. Advisor: Ole J. Forsberg. External examiner: Michael Clemens (Universal Audio)

Publications

V. Chugh and Y.-T. Yeh, "Mix-JEPA: A Joint Embedding Predictive Architecture for Music Mixing Representation," under review at *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2027.

V. Chugh, A. Phan, and K. Fatima, "Oh Sheet!: A Browser-Based Optical Music Recognition Tool for Classical and Handwritten Jazz Lead Sheets," in *Extended Abstracts for the Late-Breaking Demo Session of the 27th International Society for Music Information Retrieval Conference (ISMIR)*, 2026.

V. Chugh, "Prompt-Based Music Discovery: A Prototype Using Source Separation and LLMs," in *Extended Abstracts for the Late-Breaking Demo Session of the 26th International Society for Music Information Retrieval Conference (ISMIR)*, 2025; also presented at the *NeurIPS 2025 Workshop on Artificial Intelligence for Music: Where Creativity Meets Computation (AI4Music).*

Research Experience

Research Intern June 2026 – Sep. 2026

Audio Information Research Lab, University of Rochester | Advisor: Zhiyao Duan | HARP | Link

- Deployed 25 of HARP's 75 models in 3 months by pairing LLM-agent judgment with deterministic verification gates (dead-code pruning, preflight checks, live inference probing), cutting deployment time from hours to minutes
- Contributed to a validation framework that tests models across varied input lengths, sample rates, channel counts, and file types (95% of Spaces passing), and reviewed two 4,000+ line PRs, including the framework itself
- Built a JUCE C++ text-output component, extending support beyond audio and MIDI files

Project Lead Mar. 2026 – June 2026

Knox College | Oh Sheet!: Optical Music Recognition Web App | Link

- Built a browser-based sheet music transcription app that converts user images into editable, playable MusicXML files, reaching 40+ registered users and 300+ model calls in 2 months at $0 hosting cost
- Engineered a serverless GPU inference pipeline on Modal (A10G) with dynamic routing across a classical LLM, CNN, and jazz-specific YOLO+sequence model by score type, keeping per-call cost to ≈$0.01
- Identified and integrated a 2025 ISMIR jazz transcription model after user interviews revealed systematic failure on handwritten lead sheets, enabling a capability absent from competing browser-based tools

Knox Summer Scholar June 2025 – Aug. 2025

Knox College | Advisor: Ole J. Forsberg | Whatcha Lookin' For: Prompt-Based Recommendations | Link

- Built a full-stack natural-language music recommendation system (FastAPI, Groq-hosted LLaMA 3.3 70B, SQLite) that translates prompts like "metal songs with violin" into database queries via schema-aware prompt engineering for safe LLM-driven SQL generation
- Developed a scalable ETL pipeline, including BeautifulSoup web scraping, Demucs-based source separation, API integration, and librosa heuristics to extract instrument-presence features for 650+ tracks

Professional Experience

Data Scientist / Backend Intern Aug. 2025 – Sep. 2025

EdPi Innovate | Link

- Built the company's single-source, configurable meeting transcription and summarization system using Whisper, FastAPI, and SpeechBrain, reducing documentation time by 80% and improving organizational knowledge capture
- Engineered a speaker identification pipeline using diarization, voice embeddings, and cosine-similarity matching against employee voice profiles, enabling 95%+ accurate, fully automated speaker labeling for 50+ meetings/month
- Developed an LLM-driven summarization and action-item extraction module that maps tasks to emails, improving follow-through by 60% and reducing admin workload by 3–4 hours/week through automated follow-up processing

Data Analyst Intern Nov. 2024 – Jan. 2025

Knox College, Office of Advancement

- Analyzed 50+ years and $50M+ in alumni donation data using non-parametric statistical tests and inflation-adjusted donor metrics to evaluate cohort and involvement-based differences, informing the 2025 segmentation strategy

Data Management Intern Nov. 2023 – Jan. 2024

Joe Moore Strategies, LLC

- Designed a Google Sheets tracking system for 30,000+ clients by position, engagement level, and relationship to the company, migrating 1,500+ government-official contacts to Monday.com, reducing data retrieval time by 60%

Projects

Statistical Hypothesis Testing | *Monte Carlo Study of Normality Tests* | Link Mar. 2025 – June 2025

- Showed via parameter sweeps that Shapiro–Wilk holds nominal significance levels across sample sizes while Anscombe–Glynn under- and over-rejects, and that Shapiro–Wilk better detects asymmetry under heavy tails (t, χ^2 alternatives)

NBA Game Outcome Predictor | Link Jan. 2025 – Mar. 2025

- Engineered features from 20+ game statistics and tuned Logistic Regression, Random Forest, XGBoost, and neural network models with GridSearchCV while preventing data leakage, reaching up to 64% accuracy; documented in a 13-page paper

Talks & Presentations

"HARP: Bridging Cutting-Edge Music AI Research with Music Production Workflows," Poster, Speech and Audio in the Northeast (SANE), MIT Oct. 2026

"Automatic Mixing, FX Representations, and the WIMIR Mentee Experience," WIMIR June 2026

"Google, Recommend Some Death Metal with Acoustic Guitar: Prompt-Based Music Discovery with Source Separation and LLMs"

HORIZONS Research Symposium, Knox College May 2026
Stanford Deep Learning for MIR Summer Workshop (instructors: Iran Roman, Kitty Shi) Aug. 2025
Knox Summer Scholars, Knox College July 2025

Honors & Awards

Election to Phi Beta Kappa (sole junior inductee) 2026
College Marshal 2026
Lucile Sudbury Prize, Best Scholarly Paper in Music Theory/Analysis ($1,000) 2026
Richter Memorial Fund ($3,000; $1,200; $1,000, $900; $750) 2023–2026
ISMIR Registration Waivers ($250; $300); NeurIPS Workshop Student Grant ($200) 2025–2026
ASA DataFest Awards (Best Insight; Best Use of External Data; Honorable Mention) 2024–2026
Trustees Distinguished Scholarship ($31,000/year, 4×) 2023–2026
Knox Summer Scholars Program ($5,000; largest research grant at Knox) 2025
Clark Mills Carr Prize in Mathematics (1st, college-wide competition) 2025
Sigma Xi Best Student Seminar ($100) 2025

Teaching, Mentoring & Service

Tutoring Captain, Knox College Jan. 2024 – Present
Most-booked math tutor; 800+ hours of 1:1 tutoring in Calculus I–III, Linear Algebra, Discrete Mathematics, Introduction to Statistics, Mathematical Statistics I–II, Data Mining, and Linear Models

Research Ambassador, Knox College Sep. 2025 – Present
Advised 40+ student researchers on grant applications through 1:1 proposal reviews; supported classroom visits and events promoting undergraduate research

Teaching Assistant, Linear Algebra, Knox College Jan. 2026 – Mar. 2026
Guided 20+ students, graded 200+ assignments, and held 6+ hours of office hours weekly

Musical Background

Independent Artist | *1shh* | Link Sep. 2019 – June 2022

- Produced, mixed, and mastered original tracks released on major streaming platforms, leading end-to-end production from ideation and recording through cover art, videography, and promotion to reach 13,000+ YouTube views in 10+ countries
- Performed live with Arijit Anand (200K+ monthly listeners) at a concert with 500+ attendees

Tabla, 6th Year Examination, Prayag Sangeet Samiti Mar. 2018

Drums, Level 1 Award for Grade 3 Rock & Pop Drums, Trinity College London Mar. 2017

Technical Skills

Languages: Python, C++, R, SQL
ML & Audio: PyTorch, librosa, Demucs, Whisper, SpeechBrain, scikit-learn, TensorFlow, XGBoost
Audio Software & Production: JUCE, Max/MSP, Logic Pro, FL Studio
Tools & Infrastructure: Modal, Hugging Face Spaces, FastAPI, Git, LaTeX, SQLite